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FROM DESKTOP TO IMMERSIVE VR

In Nursing Education: Design-Based Research on System Adaptation

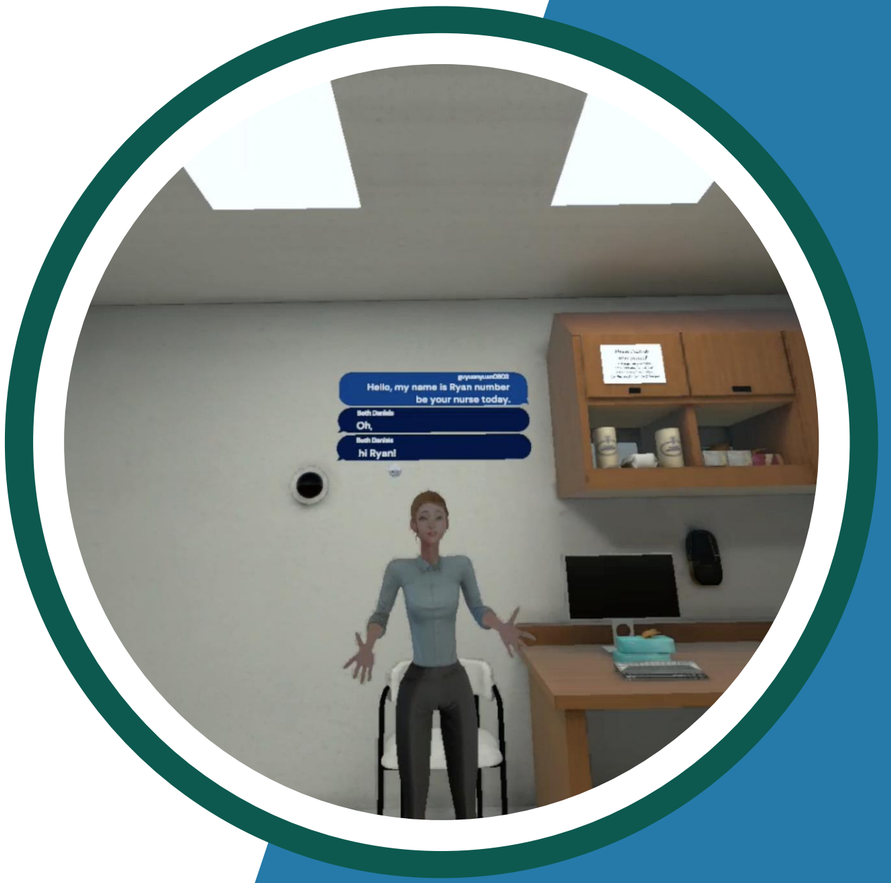
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THE BRIDGE

WHAT MUST CHANGE?

Desktop simulation scales. Immersive, AI-supported simulation offers richer practice – and asks more of faculty, IT and the institution. What must change around the technology for the richer version to last?



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WHAT WE'LL COVER

1. **Problem and questions**

   Four barriers, one tension between scale and fidelity, three questions.

2. **How we compare**

   Five DBR characteristics × three system levels; reported vs. planned evidence.

3. **Two clinics**

   Desktop VR (2022–2023) and AI-immersive VR (2024–2025); what each showed.

4. **What changed**

   The comparison matrix, then three rows up close.

5. **Clinic 3.0**

   A proposed feedback loop: empathy cues, LMS-embedded assessment, workload.

6. **What it means**

   For programs, for design researchers – and what we cannot yet claim.

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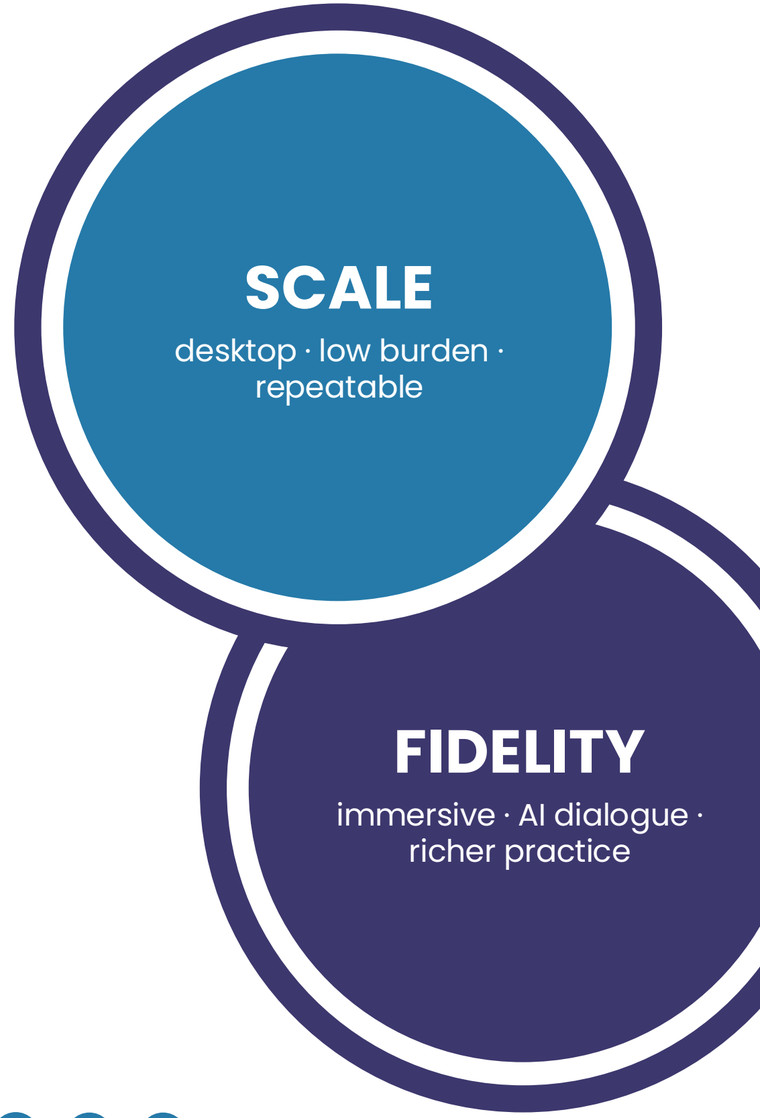
THE PROBLEM

Placement capacity
Clinical placements are limited, so practice conversations with patients are scarce.

Faculty workload
Supervising and debriefing every encounter does not scale with the cohort.

Cost of assessment
Assessment with human standardized patients – trained actors who play patients – is expensive.

Scripted patients
Branching scripts and keyword triggers limit natural interviewing and clinical reasoning.



THREE QUESTIONS

RQ1 · DIFFERENCES

How do Clinic 1.0 and Clinic 2.0 differ across van den Akker et al.'s (2006) five design-based research characteristics?

RQ2 · CONSTRAINTS AND ENABLERS

What micro-, meso- and macro-level constraints and enablers influenced implementation and scaling across the two iterations?

RQ3 · DESIGN PRINCIPLES

What design principles for AI-enhanced VR clinical communication support sustainable fidelity?

Sustainable fidelity = instructionally meaningful and organizationally adoptable.

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HOW WE ARE COMPARING THEM

DESIGN-BASED RESEARCH

Five DBR characteristics (van den Akker et al., 2006): interventionist – a real intervention; iterative – design, test, revise; process-oriented – how and why it works; utility-oriented – practical for users; theory-oriented – grounded in theory.

THREE SYSTEM LEVELS

Micro · learner: interaction and assessment evidence. Meso · faculty: orchestration and curricular alignment. Macro · institution: accreditation expectations, resource allocation, infrastructure readiness.

EVIDENCE, BY STATUS

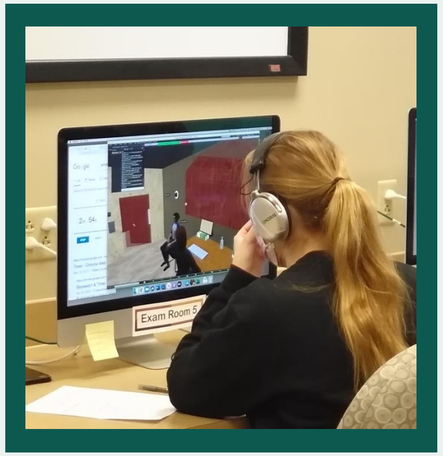
Reported: learner outcomes (self-efficacy, usability) and both clinics' design and implementation record, synthesized in the matrix. Planned: faculty workflow indicators (review time, decision latency); stakeholder interviews.

TWO CLINICS, ONE SKILL SET

CLINIC 1.0 Desktop VR · 2022–2023



Virtual patient



Learner

Computer lab

Keyboard and mouse · **types questions**

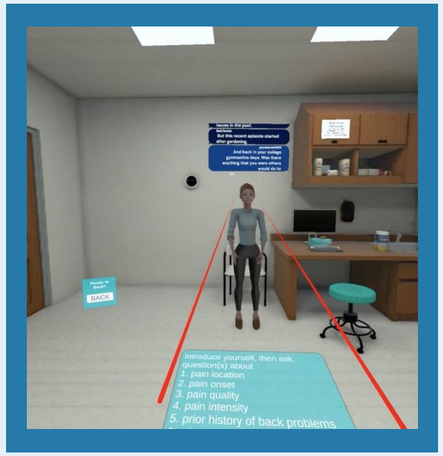
Fixed script · **replies by keyword**

SETTING

STUDENT

PATIENT

CLINIC 2.0 AI-immersive VR · 2024–2025



Virtual patient



Learner

Spatial classroom

VR headset · **speaks questions**

Large language model · **open-ended replies**

Clinic 2.0 here covers both AI-driven versions; Clinic 3.0 is the planned next generation.

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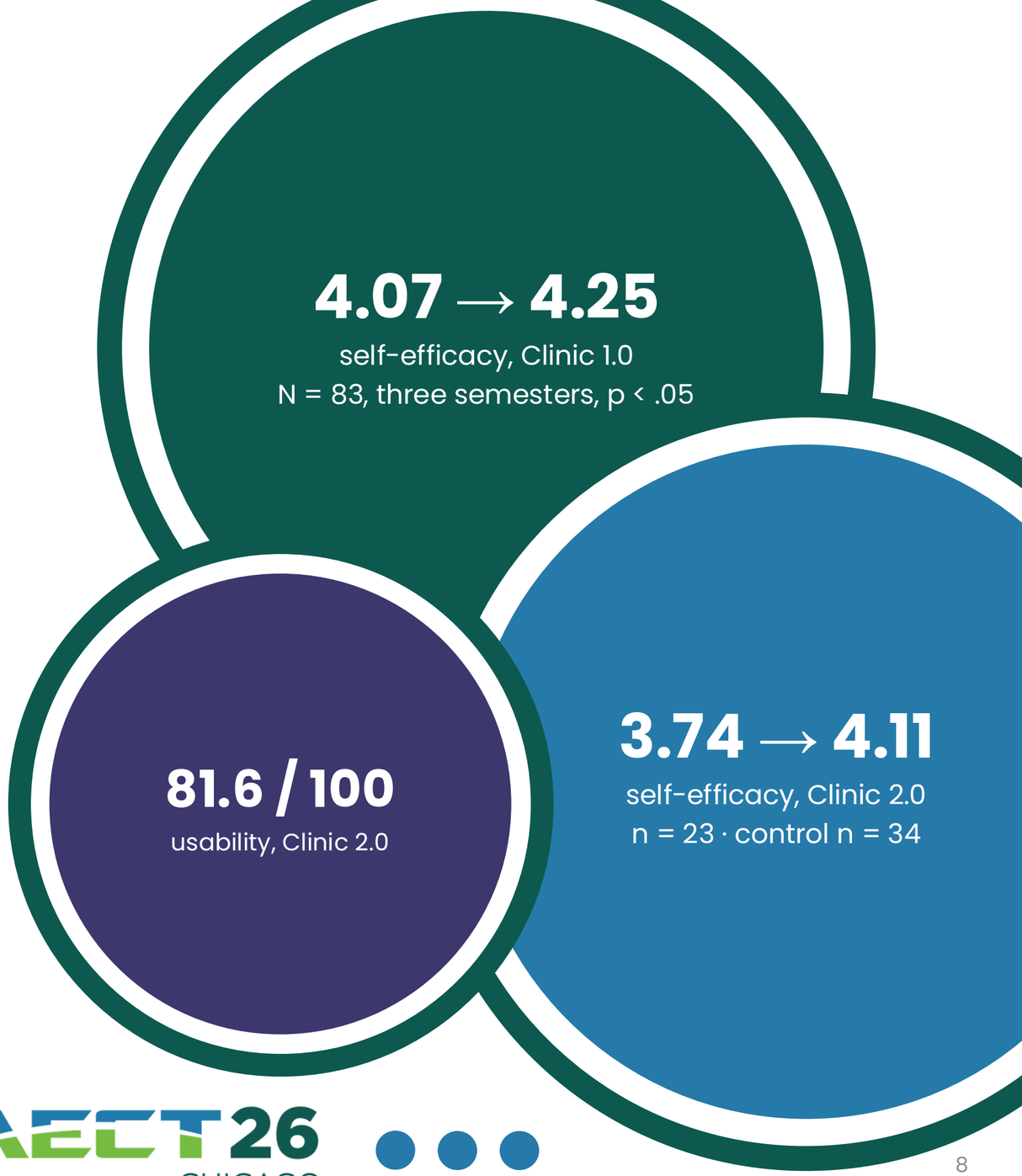
WHAT EACH CLINIC SHOWED

Clinic 1.0 – reported self-efficacy increased; activity patterns suggested students received information more than they interviewed.

Clinic 2.0 – more information-seeking: students asked follow-up questions and steered the interaction. They also reported an empathy gap in the AI patient.

The other side – Clinic 2.0 came with more orchestration, technical support and instructor (researcher) review load.

Pre/post self-efficacy from two separate studies and cohorts – not a test of which clinic teaches more.



WHAT CHANGED, LEVEL BY LEVEL

Our team's interpretive synthesis of both implementations: five DBR characteristics (van den Akker et al., 2006) × three system levels.
Clinic 2.0 column: **green** = gain at the learner level **amber** = new demands on faculty and institution

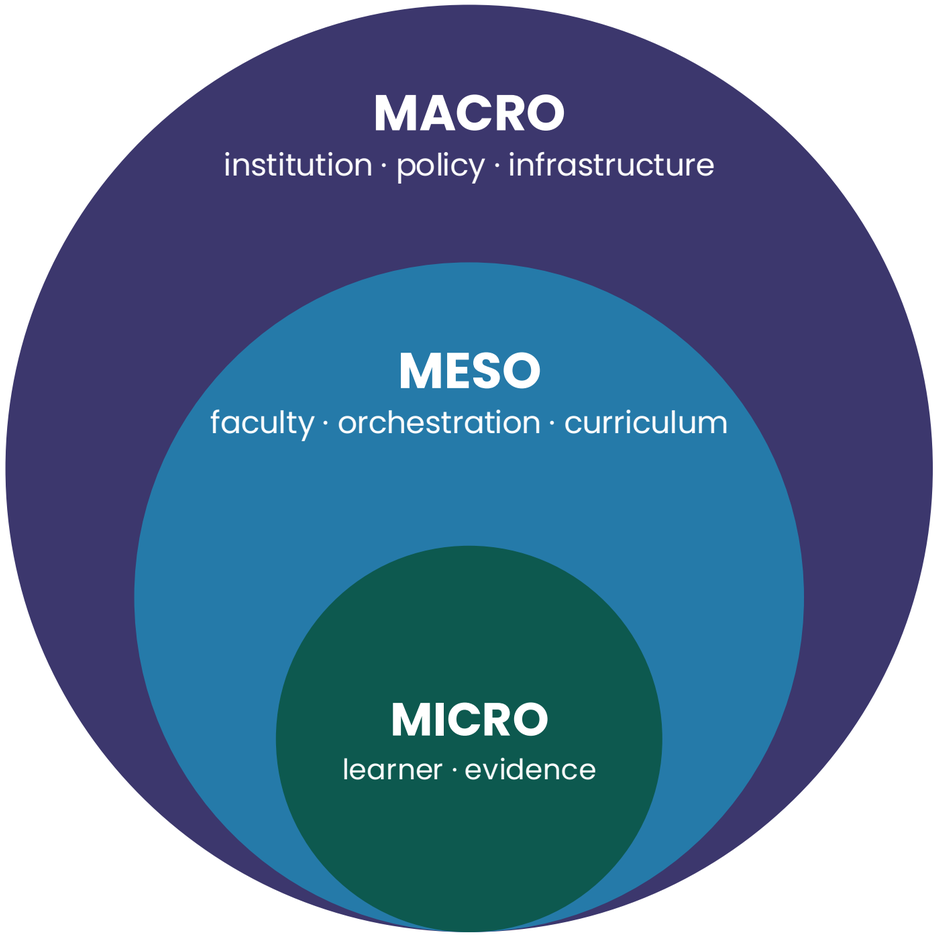
DBR characteristic	System level	Clinic 1.0 · desktop VR	Clinic 2.0 · AI-immersive VR
Interventionist	**Micro · learner**	Structured practice of encounter steps; **limited conversational agency**	Open-ended interviewing; **more learner agency** in questioning and steering
	Meso · faculty	**Fits lab scheduling** and routine instruction	Needs **new facilitation routines** and support coordination
	Macro · institution	**Minimal institutional change** needed to scale	Institutional capacity becomes a **boundary condition for scaling**
Iterative	**Micro · learner**	Iterations refine scenario flow, prompts, navigation, response triggers	Iterations refine **AI persona prompts and guardrails**, turn-taking, dialogue realism
	Meso · faculty	Iterations refine instructions, pacing, debrief routines	Iterations refine **implementation workflow and support practices**
	Macro · institution	Scaling is mainly **scheduling and lab access**	Scaling surfaces **infrastructure planning and staffing needs**
Process-oriented	**Micro · learner**	In-system behavior and pre/post measures; few natural-language traces	**Spoken dialogue traces**, interaction patterns, learner feedback on realism and empathy
	Meso · faculty	**Fewer breakdown points**; instructor focus on completion and performance	**More breakdown points** (devices, space, connectivity); orchestration and review load
	Macro · institution	Scalability with **limited systemic friction**	**Adoption friction** affecting sustainability and equity
Utility-oriented	**Micro · learner**	Self-efficacy gain supports foundational readiness	High usability and self-efficacy gain support **richer interviewing practice**
	Meso · faculty	High utility: **low overhead, repeatable**	Mixed utility: higher instructional value, **greater workload and support needs**
	Macro · institution	Strong fit for **broad deployment with minimal investment**	Fit **depends on resources, infrastructure and ongoing support**
Theory-oriented	**Micro · learner**	Desktop simulation builds confidence and fluency but **can constrain authentic agency**	AI + immersion **increases agency**; the next challenge is sustainable fidelity
	Meso · faculty	Simplicity supports routine curricular embedding	Higher fidelity **requires workflow supports** for assessment and review
	Macro · institution	Low-burden tools scale quickly but **limit fidelity**	High-fidelity tools **trigger systemic requirements** across policy, infrastructure, curriculum

THREE ROWS, UP CLOSE

	Clinic 1.0 · desktop VR	Clinic 2.0 · AI-immersive VR
THE GAIN Interventionist · micro	Structured practice of encounter steps; limited conversational agency	Open-ended interviewing; more learner agency in questioning and steering
THE COST Utility-oriented · meso	High utility: low overhead, repeatable	Mixed utility: higher instructional value, greater workload and support needs
THE BOUNDARY Theory-oriented · macro	Low-burden tools scale quickly but limit fidelity	High-fidelity tools trigger systemic requirements across policy, infrastructure, curriculum

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THE PATTERN



Fidelity is an organizational achievement, not a device property.

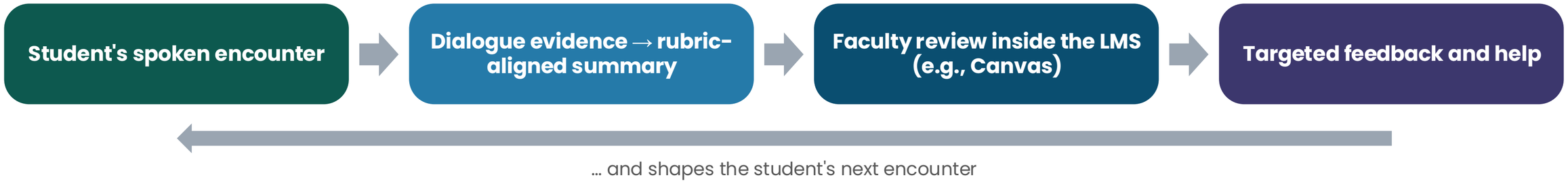
In these two iterations, each gain at the micro level came with new demands at the meso and macro levels.

Clinic 1.0 was easy to deploy, but its fidelity was limited. Clinic 2.0 raised fidelity – and pulled facilitation, infrastructure and policy into the design problem.

A design proposition from two cases, not a law.

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CLINIC 3.0: THE NEXT CYCLE



1

Close the empathy gap

Multimodal cues – facial expression, posture – aligned with clinical reasoning needs, answering the empathy gap students reported.

2

Make assessment coherent

LMS integration (e.g., Canvas): rubric-aligned summaries reviewed in existing grading workflows; analytics show which skills each student still needs to improve.

3

Test the feedback loop

Does instructor decision time fall and feedback specificity rise, relative to Clinic 2.0's manual review, with student outcomes maintained or improved?

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WHAT THIS MEANS

For nursing programs: a change plan

New facilitation and debrief routines for AI dialogue · staffing and support capacity for devices, space and connectivity · assessment review inside the LMS grading workflow, not manual review of every encounter.

Three early principles (RQ3), for Clinic 3.0 to test

Match fidelity to the support capacity you have · design the review workflow with the simulation, not after it · treat policy, infrastructure and curriculum as design variables, not background.

For Systems Thinking and Change

The beginnings of a systemic adoption framework: the matrix as a reusable template for seeing where an innovation's demands may arise, plus the feedback-loop mechanism Clinic 3.0 tests.

What we cannot yet claim

Two iterations, different cohorts and years; a single team's interpretive synthesis. The comparison describes systemic adaptation; it does not isolate causes.



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